IGCSE 0607 Practice Questions — E1.2

Sets

Total Questions: 15 — **Total Marks:** 91 Marks

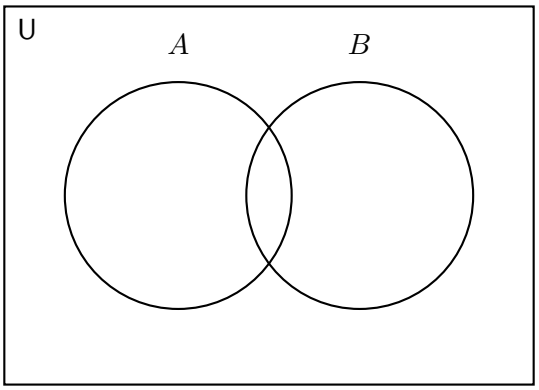
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  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

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1 The universal set U contains all integers from 1 to 10 inclusive.
Set A contains all the even numbers in U.
Set B contains all the prime numbers in U.

(a) Complete the Venn diagram to show the elements of these sets.



[2]

(b) List the elements of $A \cap B$.

..... **[1]**

(c) Find $n(A \cup B)'$.

..... **[1]**

2 $\mathsf{U} = \{x : x \text{ is an integer and } 1 \leq x \leq 10\}$
$A = \{x : x \text{ is a prime number}\}$
$B = \{x : x \text{ is a factor of } 12\}$

(a) List the members of set A.

..... **[1]**

(b) List the members of set B.

..... **[1]**

(c) Find $n(A \cup B)$.

..... **[1]**

(d) Write down the elements of $A \cap B$.

..... **[1]**

(e) Write down a number that is in U but is not in $A \cup B$.

..... **[1]**

3 The universal set U and sets A and B are defined as follows:

- $\mathsf{U} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- $A = \{2, 3, 5, 7\}$
- $B = \{1, 3, 5, 7, 9\}$

(a) List the elements of $A \cap B$.

..... **[1]**

(b) Find $n((A \cup B)')$.

..... **[2]**

(c) Write down the correct set notation symbol $(\in, \notin, \subset, \cup, \cap)$ to complete the statement:

$2 \ldots A$

[1]

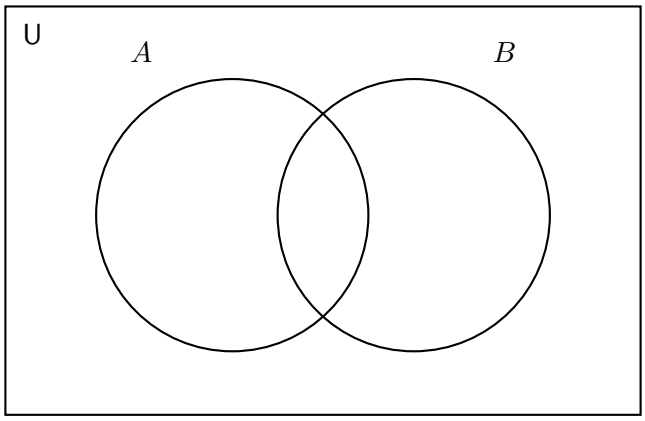
(d) Write down the correct set notation symbol to complete the statement:

$\{1, 9\} \ldots B$

[1]

4 Let U be the set of integers from 1 to 10 inclusive.
Let A be the set of prime numbers.
Let B be the set of odd numbers.

(a) Complete the Venn diagram to show the elements of these sets.



[2]

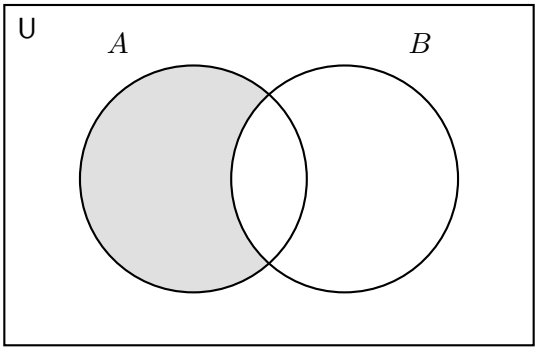
(b) List the elements of $A \cap B'$.

..... **[1]**

(c) Find $n(A \cup B)$.

..... **[1]**

(d) Write down the set notation for the shaded region in the Venn diagram above.



..... **[1]**

5 The universal set U and sets P and F are defined as follows:

- $\mathsf{U} = \{x : x \text{ is an integer} 1 \leq x \leq 10\}$
- $P = \{x : x \text{ is a prime number}\}$
- $F = \{x : x \text{ is a factor of } 18\}$

(a) Write down the elements of set F.

$F = \{$ ..... $\}$ **[1]**

(b) List the elements of $P \cap F$.

$P \cap F = \{$ ..... $\}$ **[1]**

(c) Find $n((P \cup F)')$.

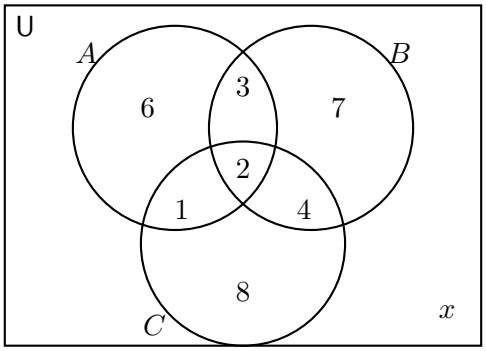
..... **[2]**

(d) Write down the correct set symbol, $\in$ or $\notin$, to make this statement true:

$5 \ldots\ldots F$

[1]

6 The Venn diagram shows the number of elements in each region for sets A, B, and C.



(a) Given that $n(\mathsf{U}) = 35$, find the value of x.

$x =$ ..... **[2]**

(b) Find $n(A \cap C')$.

..... **[1]**

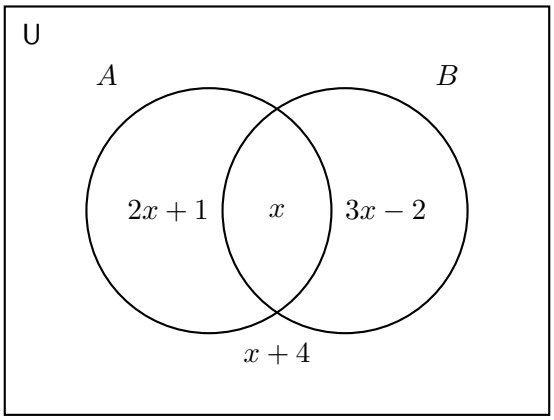
(c) Use set notation to describe the set of elements whose number is 4.

..... **[1]**

(d) Find $n(A \cup (B \cap C))$.

..... **[2]**

7 The Venn diagram shows the number of elements in sets A and B, in terms of x.
U is the universal set and $n(\mathsf{U}) = 52$.



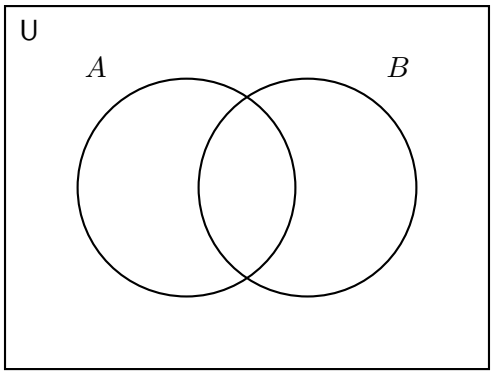
(a) Form an equation in x and solve it.

$x =$ ..... **[3]**

(b) Find $n(A \cup B')$.

..... **[2]**

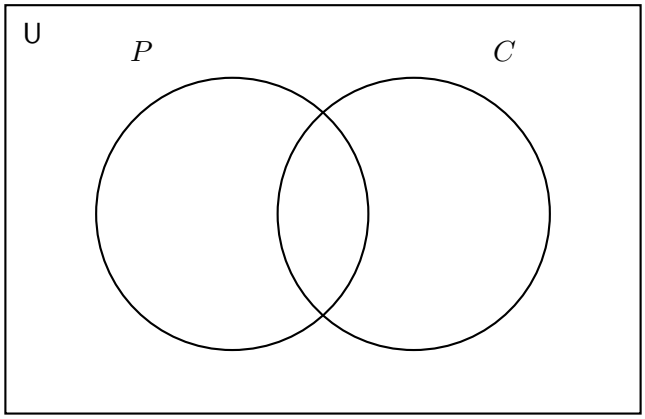
(c) On the Venn diagram below, shade the region $A \cap B'$.



[1]

8 There are 45 students in a class.
Let U be the set of students in this class.
Let P be the set of students who study Physics.
Let C be the set of students who study Chemistry.
$n(P) = 28$
$n(C) = 22$
$n(P \cap C) = x$
$n((P \cup C)') = 4$

(a) Complete the Venn diagram in terms of x.



[2]

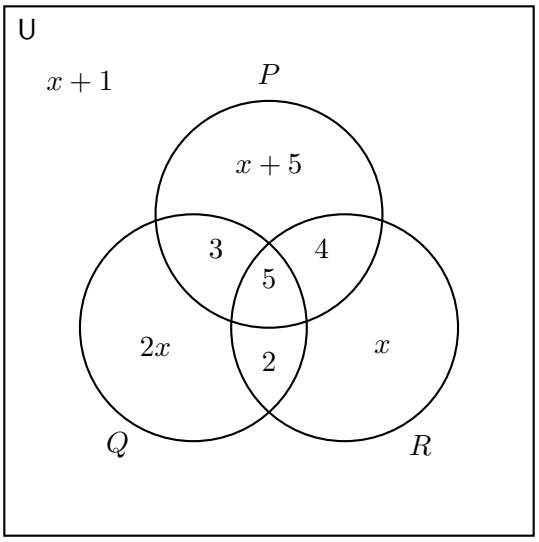
(b) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[2]**

(c) Find $n(P \cap C')$.

..... **[1]**

9 The Venn diagram shows the number of elements in each region for the sets P, Q and R.



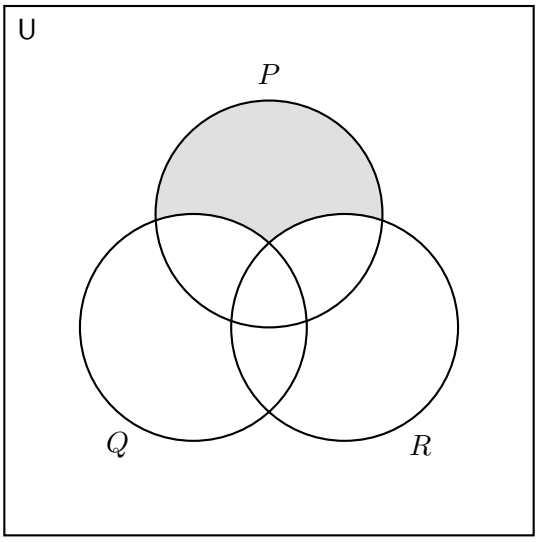
(a) The total number of elements in the universal set is 45.
Form an equation in terms of x and solve it to find the value of x.

$x =$ ..... **[3]**

(b) Find $n(P \cup Q)$.

..... **[1]**

(c) On the Venn diagram below, shade the region $P \cap (Q \cup R)'$.

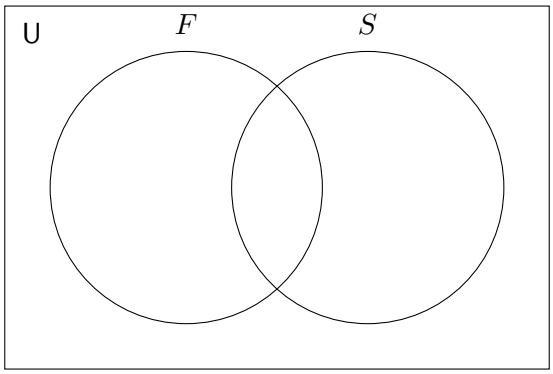


[1]

10 In a class of 40 students:

- 25 study French (F)
- 18 study Spanish (S)
- 5 study neither French nor Spanish

(a) Complete the Venn diagram to show this information.



[3]

(b) Find $n(F \cap S')$.

..... **[1]**

(c) A student is chosen at random from the class.
Find the probability that they study exactly one of the two languages.

..... **[2]**

(d) Two students are chosen at random from those who study French.
Find the probability that both also study Spanish.

..... **[1]**

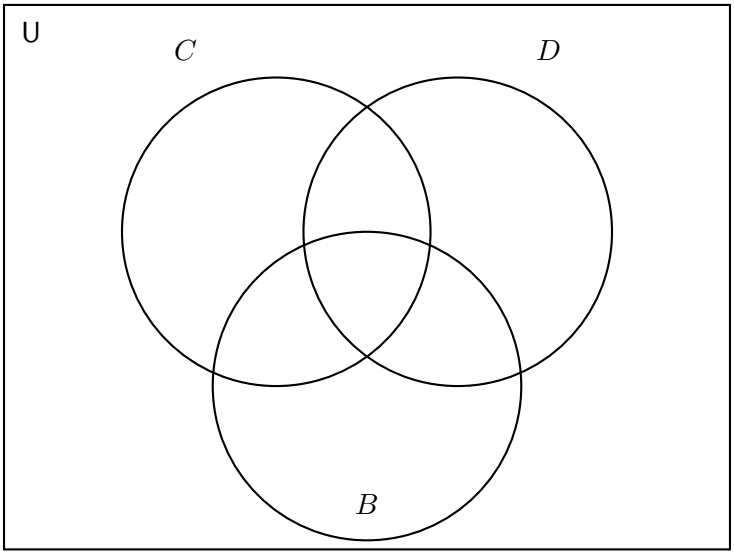
11 In a survey of 100 people, information was collected about the pets they own.

Let U be the set of people surveyed.
Let C be the set of people who own a cat.
Let D be the set of people who own a dog.
Let B be the set of people who own a bird.

- 8 people own all three pets.
- 12 people own a cat and a dog but not a bird.
- 7 people own a cat and a bird but not a dog.
- 10 people own a dog and a bird but not a cat.
- 3 people do not own any of these pets.

The number of people who own only a cat is x.
The number of people who own only a bird is also x.
The number of people who own only a dog is $2x$.

(a) Complete the Venn diagram to show this information.



[3]

(b) Find the value of x.

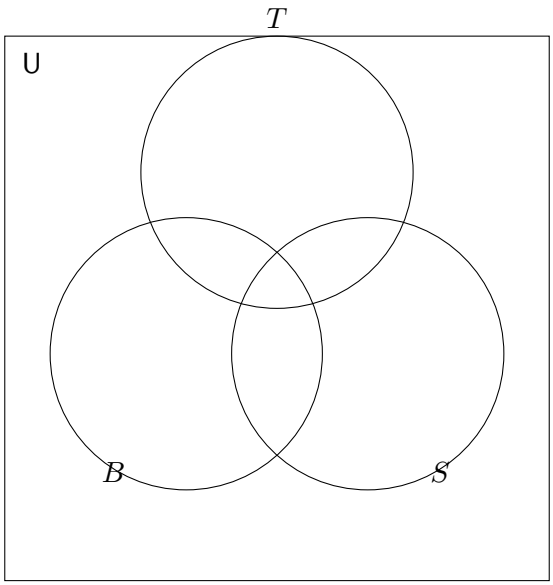
$x =$ ..... **[2]**

(c) Find $n((C \cup D) \cap B')$.

..... **[2]**

12 There are 60 people in a sports club.

- 30 play Tennis (T)
- 28 play Badminton (B)
- 25 play Squash (S)
- 12 play both Tennis and Badminton
- 9 play both Tennis and Squash
- 11 play both Badminton and Squash
- 4 play all three sports



(a) Complete the Venn diagram.

[3]

(b) Find $n(T \cup B \cup S)'$.

..... **[1]**

(c) Find $n(T \cap (B \cup S))$.

..... **[1]**

(d) Use set notation to describe the region containing the number 9 (the number of people who play Badminton only).

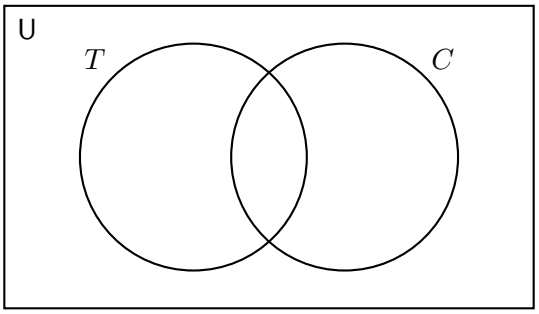
..... **[2]**

13 In a sports club, the universal set U represents all the students.
The set T represents students who play Tennis.
The set C represents students who play Cricket.

You are given the following information:

- $n(\mathsf{U}) = 50$
- $n(T) = 3x$
- $n(C) = 4x$
- $n(T \cap C) = x$
- $n((T \cup C)') = 2$

(a) Complete the Venn diagram, writing the number of students in each region in terms of x where appropriate.



[3]

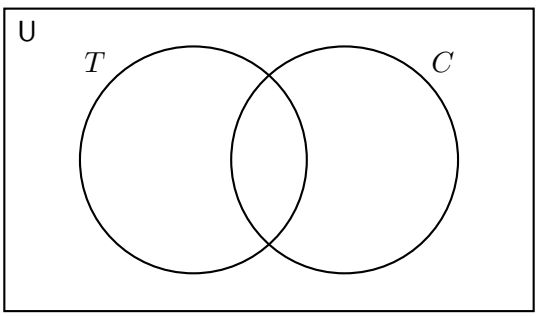
(b) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[2]**

(c) Find the number of students who play Tennis but not Cricket.

..... **[1]**

(d) On the Venn diagram below, shade the region $T' \cup C$.

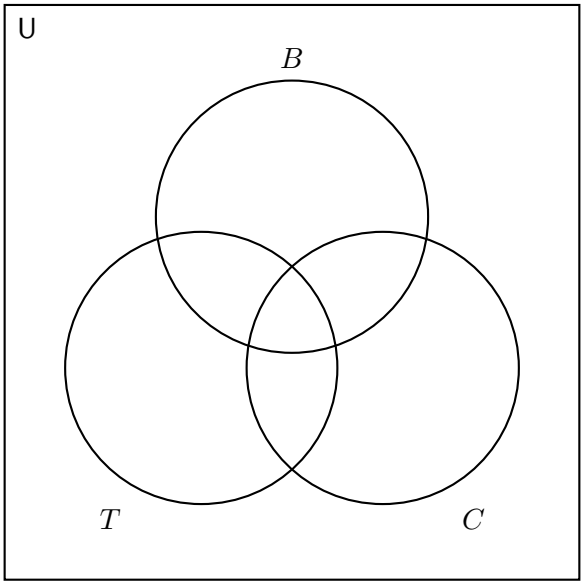


[2]

14 In a survey of 103 people, the following information was recorded about their use of three types of transport: Bus (B), Train (T), and Car (C).

- 15 people use all three types.
- 28 people use both a Train and a Car.
- 22 people use both a Bus and a Car.
- 30 people use both a Bus and a Train.
- x people use only a Bus.
- The number of people who use only a Train is twice the number of people who use only a Bus.
- The number of people who use only a Car is 5 more than the number of people who use only a Bus.
- 8 people do not use any of these types of transport.

(a) Complete the Venn diagram in terms of x to show this information.



[3]

(b) Find the value of x.

$x =$ ..... **[2]**

(c) Find $n(T \cup C)$.

..... **[1]**

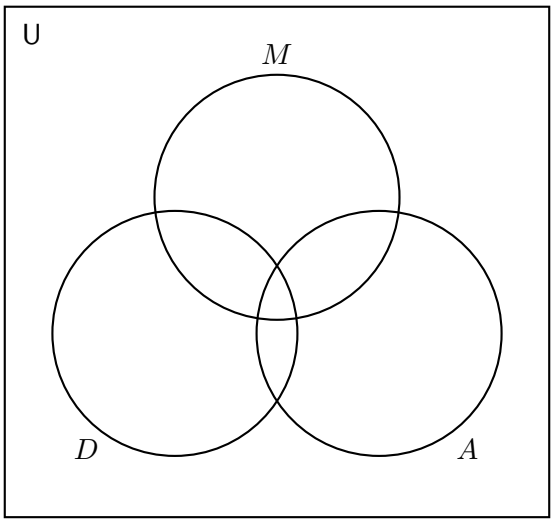
(d) A person is chosen at random from those who use a Train.
Find the probability that this person also uses a Car but does not use a Bus.

..... **[2]**

15 In a year group of 120 students, they can choose to study Music (M), Drama (D), and Art (A).

- 12 students study all three subjects.
- 25 students study Music and Drama.
- 32 students study Drama and Art.
- 28 students study Music and Art.
- 50 students study Music.
- 65 students study Drama.
- x students study Art only.
- 15 students study none of these three subjects.

(a) Use this information to complete the Venn diagram.



[3]

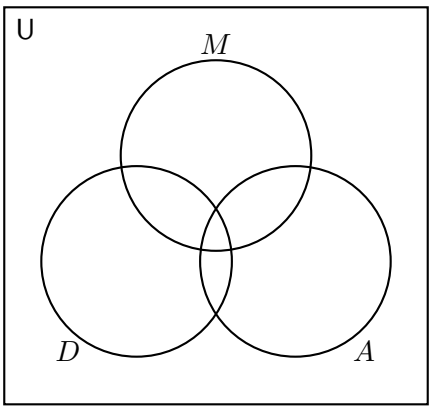
(b) Find the value of x.

$x =$ ..... **[2]**

(c) Find $n(M \cup (D \cap A))$.

..... **[2]**

(d) On the Venn diagram below, shade the region $M \cap (D \cup A)'$.



[1]